Seaglider Rev. E Hardware Installation

rev 1.0, 05-Sep-2023

1. Critical differences with Rev B

1. When used with a reed switch for on/off, on and off are on the same side (starboard). Hold the magnet in place for approximately one second and then remove it to turn the glider on. Repeat to turn the glider off.
2. Alternatively, a shorting plug can also be used to turn the glider on and off. Appropriate endcap wiring must be in place (see below) and a shorting plug or cable must be available. Selection of reed switch vs shorting plug control of on/off is done via a software setting on the supervisor.
3. The pin out for the sensor serial ports on the motherboard have changed. Pin outs on the tailboard remain the same. On Rev E, pin outs are generally the same on main and tail, following the Rev B tail convention.
4. Many more ports are configurable for RS232/logic level and/or voltage (see below). Always confirm pin outs, serial voltage level, and power voltage level of your devices and the ports they are connecting to.
5. Rev E console connection operates at 115200 baud.
6. Both hardware and software gain are available for the external pressure sensor. Check with your service provider if you are unsure about the hardware gain configuration on your board.
7. The 3-pin altimeter serial port connector (J30 on Rev B) moves to a 4-pin connector with a different pin out (see below).
8. Battery merge for a 15V univolt system is done via a single jumper connector at J39. Jumper pins 1-2 to parallel the two battery packs.

2. Basic architecture of Rev E

The Seaglider Rev E motherboard uses an ARM microprocessor as the primary computational engine for glider functions. A second ARM processor is used for motor control functions. An 8-bit microprocessor is used for supervisor and analog input functionality. The supervisor is responsible for:

- on/off control of the glider (reed switch or shorting plug detection)
- software watchdog
- continuous real-time fuel gauging
- battery voltage monitoring
- internal sensor monitoring (temperature, humidity, internal pressure)
- external pressure sampling

The motor controller has access to an onboard accelerometer and can independently sample the external pressure sensor. These data are provided in motor move records, but are not yet used operationally. The motor controller uses a 16-bit AD converter for motor feedback potentiometers. These values are divided by 16 for compatibility/familiarity with Rev B 12-bit numbers. The only user visible effect of this is that some AD count values are now presented as floating point numbers.

Firmware for the main processor and motor controller are generally user upgradeable. Supervisor firmware is not user upgradeable. Motor controller and supervisor firmware generally changes much less frequently than the main firmware. All of the firmware modules have interdependencies and thus compatibility must be considered when making updates.

3. Available sensor ports and configuration

Standard sensor port pin out is: 1-GND, 2-TX, 3-RX, 4-power (TX and RX sense is Seaglider TX to sensor and Seaglider RX from sensor). Some ports have jumper selectable voltage: 5B (10 or 24), 5C (10 or 24), and 6 (10/15 or 3.3). Port 4 supports software selection of voltage (10 or 24) on different pins. In a univolt 15V glider, the 10/24 distinction is meaningless, but the jumpers must still be installed. GPS and compass voltage are also jumper selectable (10/15 or 3.3). **Always** confirm the device voltage input before setting the jumpers and energizing the system. Sending 15V to a 3.3V device will almost certainly damage the device beyond repair.

Three ports (3D, 5A and 6) are always logic level serial (not RS232). Several ports can be made logic level via hardware modification to the motherboard. Many boards ship standard with logic level GPS, Iridium phone and compass ports for example. Check with your service provider if you are unsure about the configuration of your board. Key indicators are the presence or absence of U27 (Iridium phone), U53 (GPS), and U54 (compass) on the board. If that part is present then the port is configured for RS232. If the corresponding part is absent, then that port is configured for logic level. Connecting a logic level device to an RS232 port will almost certainly damage the device. Likewise, connecting an RS232 device to a logic level port on the motherboard will very likely damage the motherboard.

Port	Main connector	Tail connector	Voltage jumper	Notes
2	J8	J10	None, always 10	
3D	J46	N/A	No power available	GPIO on pins 4,5
4	J15	J12	Software select (pin 4=10, pin 5=24)	
5A	J16	N/A	None, always 10	
5B	J17	J13	UART5B_PWR (1-2=10, 2-3=24)*	
5C	J18	J16	UART5C_PWR (1-2=10, 2-3=24)*	
5D	J19	J21	None, always 10	Serial tee on tailboard
6	J41	J17	UART6_FREQ2_PWR (1-2=10, 2-3=3.3)	

* By convention, jumper pins 1-2 when installing jumpers for 5B and 5C on a 15V univolt system.

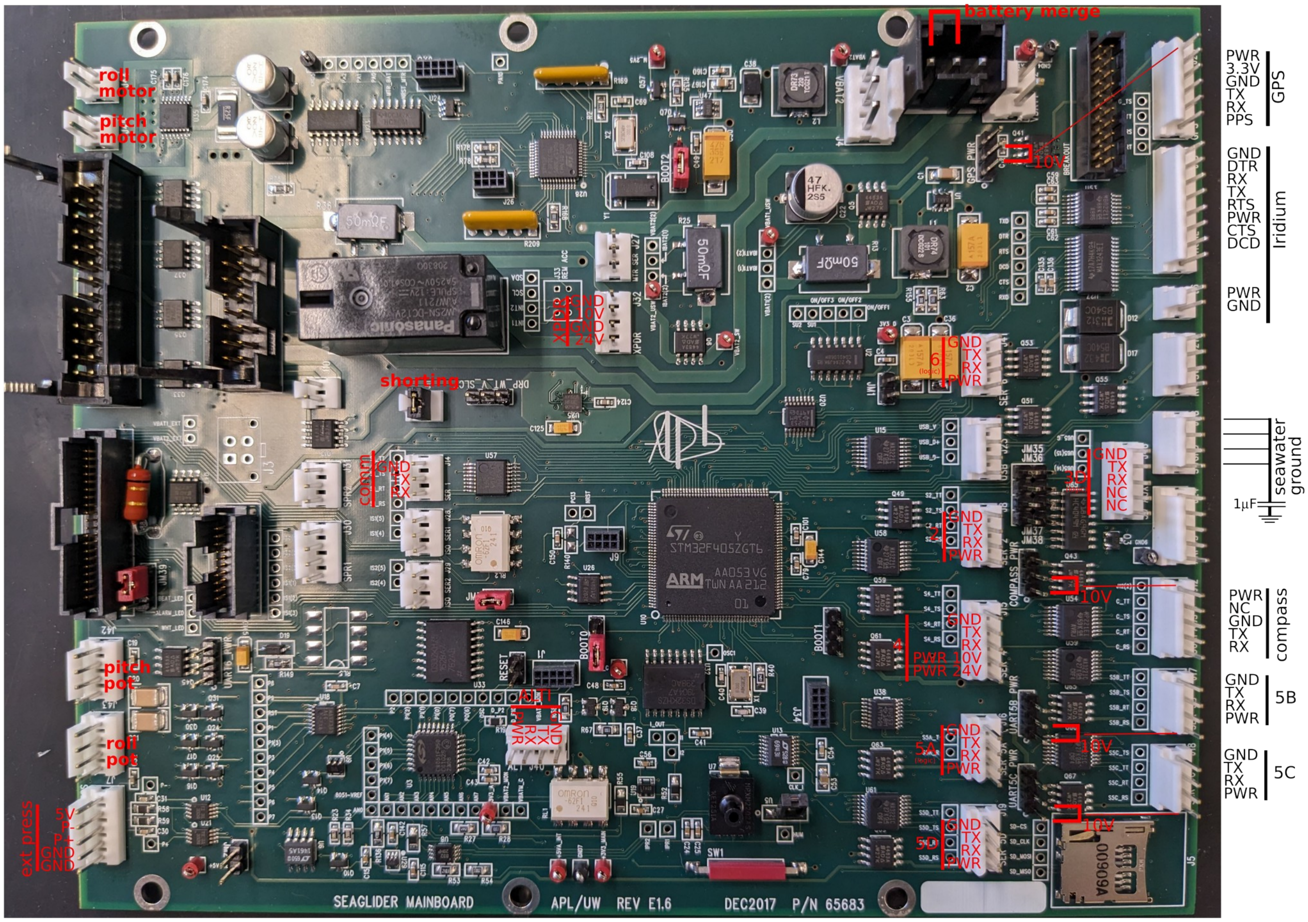


Figure 1: Seaglider Rev E.1.6 motherboard connector locations

4. Motherboard connections

Pin outs and physical location for most mainboard connections are the same on Rev B and Rev E. With appropriate care about voltage and serial level for compass, GPS, and Iridium phone as noted above, most connections can be migrated directly. The transponder serial connection is an exception to this (see below). Additionally, some old style GPS connections need special handling.

Connector	Rev B ref	Rev E ref	notes
Compass	J20	J12	
RF switch	J21	J21	
Phone power	J6	J6	
Iridium phone	J5	J13	Rev B pin 6=NC, Rev E pin 6 same as J6-1 (pwr)
GPS	J4	J20	Rev B pin 2=Vaux. Rev E pin 2=always on 3.3V
XPDR (power)	J15	J32	
Roll motor	J3 (pins 1-2)	J44	
Pitch motor	J3 (pins 3-4)	J45	
Pitch pot	J10 (pins 1-4)	J42	
Roll pot	J10 (pins 5-8)	J43	

External pressure	J19	J7	
Altimeter (XPDR serial)	J30		see below

5. Transponder connection

The recommended configuration is to connect the transponder serial to port 5A or 6. The connection must be logic level. The previous default was to connect to J40 (*ALTI* on silkscreen) which provides a pass through connection via the supervisor. This configuration still works (choose “null port” in the configurator), but will be deprecated in future hardware revisions. When connecting to 5A or 6, do not connect pin 4 (the transponder has always on power and does not use the switched port power). The Rev B pin out is 1-TX, 2-RX, 3-GND. The Rev E pin out (5A, 6, or ALTI) is 1-GND, 2-TX, 3-RX, 4-NC. Always remove the 3-pin connector and install a 4-pin connector to reduce the likelihood of incorrect connections.

6. Ribbon cables

There are up to 5 ribbon cables running fore-aft in a Rev E equipped glider. The high density 40-pin cable and the standard density 26-pin (previously “rainbow”) cable are always required. The high density 20-pin cable is required if port 5D or any of the spare connectors are used on the tailboard. The 16-pin standard density cable is generally not used on a SG variant Seaglider (it supports an aft mounted battery on SGX). A fifth cable supports an aft mounted Iridium GPS board (standard on SGX).

7. Aft endcap connections

Most endcap sensor bulkhead to tailboard connections can be migrated directly from Rev B to Rev E when moving from 4-pin to 4-pin header (optical #1 to 5B for example). If using port 4 (J12, *SERIAL #4*), remove any existing 4-pin connector from the bulkhead wiring and install a 5-pin connector to reduce the risk of incorrectly plugging the connector to the board by being off by one pin. Linpots, valve, boost and main motors, and SBE CT connections (J8 *TEMP* and J9 *COND*) also migrate directly.

The connection for the communications cable (generally bulkhead connector A) is now an 8-pin connector.

Pin	Rev B (J3)	Rev E (J3)	IE55 pin	MCBH8 pin
1	GND	GND	3 (red)	3 (red)
2	RX	RX	1 (white)	1 (white)
3	TX	TX	2 (black)	2 (black)
4	DTR/ext 10V	DTR	4 (green)	4 (green)
5	ext 24V	ext 24V	5 (blue)	5 (blue)
6	GND	GND	6 (brown)	6 (brown)
7	N/A	ext 10V	4 (green)	7 (yellow)
8	N/A	GND		8 (orange)

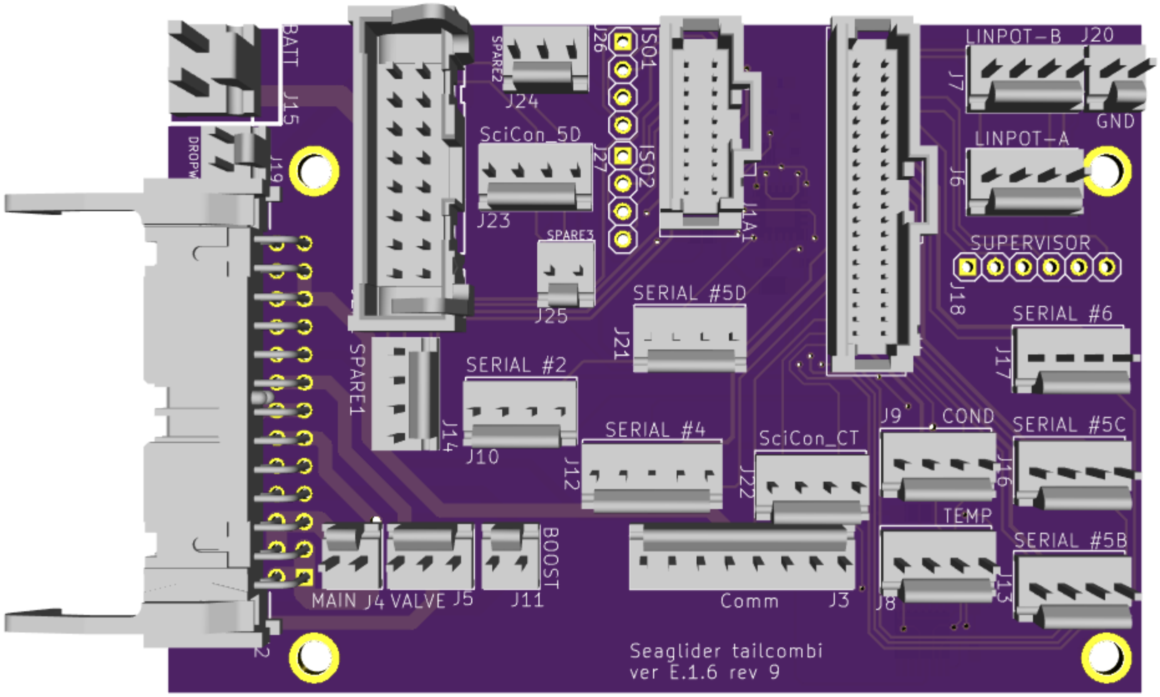


Figure 2: E.1.6 tailboard. Position of the Jx number marks pin 1.

8. Shorting plug wiring

If configured to use a shorting connection for on/off then wiring must be installed between bulkhead connector(s) and J18 (supervisor). Pin out for J18 is:

1. supervisor reset (active low)
2. on/off signal (also on pin 1 of J11 on the mainboard for testing, see Fig. 1)
3. GND (also on pin 2 of J11 on the mainboard)
4. strobe LED
5. heartbeat LED
6. alarm LED

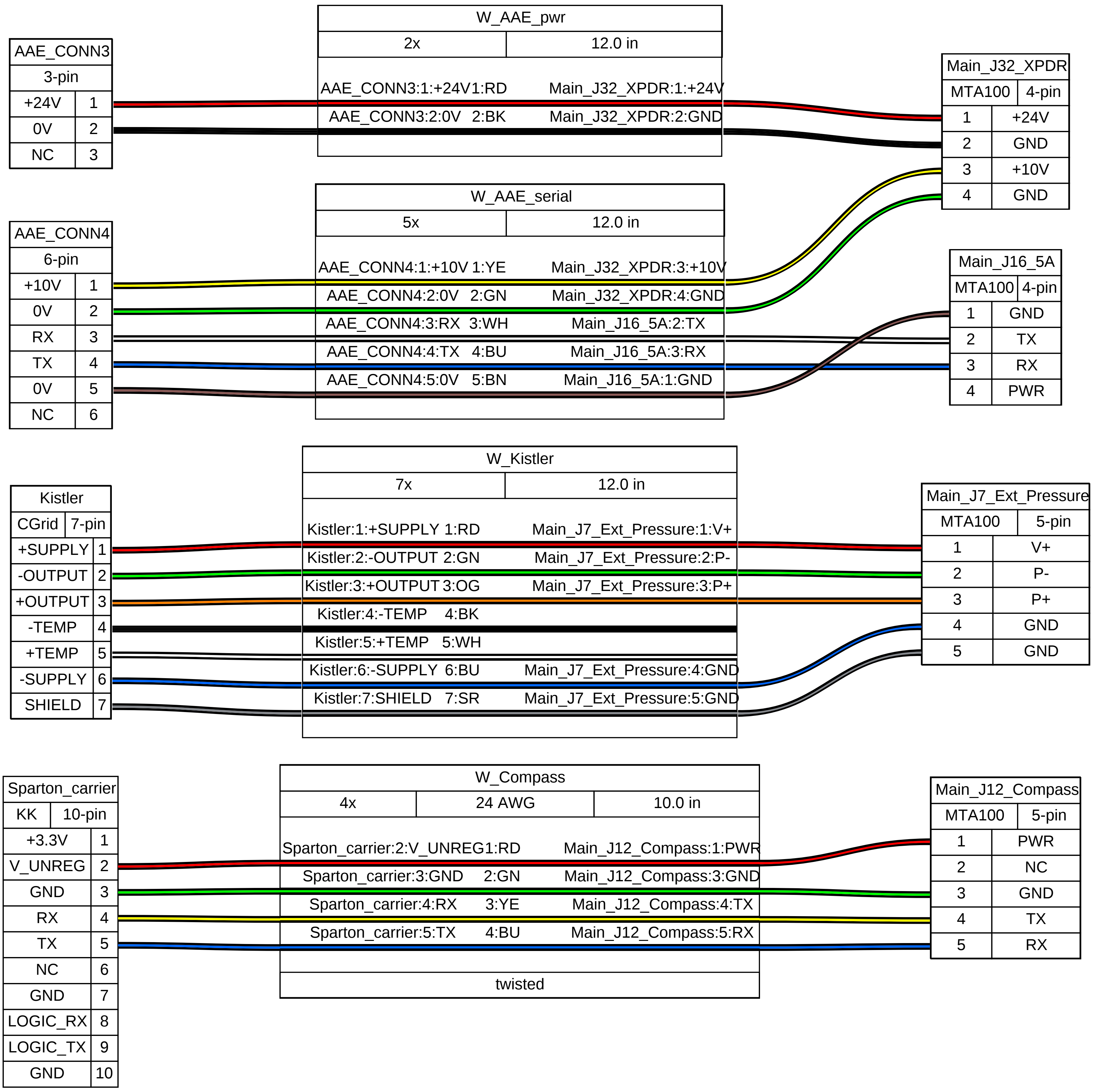
On/off is controlled by shorting pin 2 to GND (connecting pins 2 and 3). When connected the glider is on. When not connected (shorting plug or cable removed) the glider is off. The ideal wiring configuration is a potted shorting plug with embedded LEDs. A second option is a potted plug that just shorts 2-3. In these situations, a 6-pin IE55 bulkhead can be wired pin-for-pin to J18. If a shorting plug is not available, a straight through cable can be used to make a connection between two different bulkhead connectors (HII used endcap ports F and I). In this configuration connect pin 2 of endcap port F to pin 2 of J18. Connect pin 2 of endcap port I to pin 3 of J18. With a straight through IE55 cable installed between F and I, pins 2 and 3 of J18 will be connected and the glider will turn on.

Be mindful of the orientation of the connector when mating to J18. The J18 header does not have a locking tab. A standard MTA connector mated to J18 should have the wires exiting towards the top of the board (relative to the view in Fig. 2), the same as other connectors on that side of the tailboard.

9. Tailboard tee'd ports

Revision 9 of the E.1.6 tailboard supports a tee or Y connection for serial port 5D and the Sea-Bird CT port. This allows one serial sensor (e.g., RBR Legato CTD) and/or the SBE CT sail to be physically connected to both the mainboard and to the science controller. Connect the aft endcap bulkhead connector to J21 (*SERIAL #5D*) and then wire J23 (*SciCon_5D*) to a port on the science controller. Likewise, connect the SBE CT boards to J8 (*TEMP*) and J9 (*COND*) as normal and then connect J22 (*SciCon_CT*) to the science controller CT frequency counting channel. Circuits on the tailboard detect which device (glider or science controller) is powering the port and direct the digital data appropriately. Only one device should be configured to use the sensor at any time. If you do not have a science controller then use J21 (5D) and the CT ports as per normal.

Revision Date	Note



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